PyGameMaker — Tutorial 10: Turn-Based Multiplayer by File Exchange — Teacher Guide

Companion for the in-app tutorial (**Help > Tutorials > Turn-Based Multiplayer: Tic-Tac-Toe by File Exchange**, 5 pages) and the matching student handout and worksheet. Students should have finished Tutorial 4 or 6 and be comfortable with variables, conditions and the Draw event. This is the most demanding tutorial in the series and the only one that needs two computers.

Overview

Students build Tic-Tac-Toe for two computers in four phases: the board (single player), host and join, taking real turns, and winning and playing again. New ideas: **turn-based multiplayer through a shared folder** (no live connection, so it works through school firewalls), **global variables set by the network** (global.player_id, global.round_number, global.waiting_for_players), **shared variables**, **End Turn**, and the **Round Resolved** event.

> The historical note in the tutorial (play-by-mail, VGA Planets) makes a good opening: the game exchanges files exactly as players once exchanged letters.

Suggested Timing (75-90 minutes, best over two sessions)

The tutorial says 25-30 minutes for a confident student; the nine repeated cell checks, the eight win lines and the lab setup take much longer.

Segment	Time	What happens
Introduction and pairs	10 min	Explain the idea; form pairs; prepare the shared folder
Phase 1: the board	20 min	Pages 1-2; the nine repeated checks are tedious but identical in shape
Phase 2: host and join	15 min	Page 3; first real test between two computers
Phase 3: taking turns	20 min	Page 4
Phase 4: winning	15 min	Page 5; eight win lines
Worksheet / exit ticket	5-10 min	Worksheet parts A-C

Phase 1 needs no second computer. A good split: session 1 = Phase 1 (and Phase 2 if time), session 2 = the rest with the lab set up.

Lab Setup (do this before the lesson)

- **The shared folder is the whole trick.** Both computers of a pair must read and write the *same* folder. Options: a mapped network drive or UNC path on the school server (best: \server\share\tictactoe or a drive letter), or a synced folder (Dropbox, OneDrive). Give each pair its **own subfolder** (for example tictactoe_pair3) so pairs do not interfere.
- Both computers must type the **exact same folder path** in the Host and Join actions. A drive letter that means something different on each computer is a common trap; a UNC path is safest.

- The game checks the folder about **once per second**, so a move takes roughly a second (sometimes two) to appear on the other screen. With a synced cloud folder, sync delays add to that.
- **Solo testing on one computer:** use a plain folder name (tictactoe_files) and run two Test Game windows, one pressing H and one pressing J. The bundled fichier_1 sample works the same way and is a finished version students can compare with.
- The game never freezes if the folder cannot be reached: it keeps running single-player (and "Waiting for opponent..." simply stays).
- The host waits for the other player's move only up to the **round deadline** (30 seconds by default); after that a missing player is skipped for that round.

Step-by-Step Walkthrough and Common Issues

Phase 1: the board

- Everything lives on one object, obj_game. The nine cells are nine plain variables (cell_column_row), each with a fixed name, because there is no way to build a variable name from other values in this engine.
- The nine click checks and the nine draw checks are identical in shape; the tutorial's tables give the rectangles. Students often make typos in the numbers. Consider pairing students or giving the tables on paper.
- **Set draw color white first.** The tutorial draws black by default and a new room is black, so without *Set draw color* the whole board is invisible (the reference project shows nothing at all until the colour is set). The tutorial now includes that block.

Phase 2: host and join

- H hosts and becomes X; J joins and becomes O. Both computers run the same project. If both press H, both say "You are X" and nothing works.
- global.waiting_for_players stays 1 until the host has as many players as the Max players setting (2). No counting is needed.

Phase 3: taking real turns

- Round 1 is X's; odd rounds are X's, even rounds O's. The player who is *not* on turn must still submit something, so the game does not wait for the deadline: the Step event sends an empty **End Turn** once per round (the last_round_acted variable stops it from firing every frame).
- The click handler writes the mark with **Set a Shared Variable** using the cell's *literal name*, then **End Turn**. The mark appears on both screens only when the round resolves, about a second later; students think "it did not work" and click again.
- Marks are drawn from global.cell_*. A shared variable nobody has set reads as the number **0**, not empty text, so the drawing checks compare with 0.
- In the reference project, a click made out of turn is ignored, and the two computers always show the same board.

Phase 4: winning and playing again

- The **Round Resolved** event runs on both computers after each confirmed round, so both compute the winner themselves.
- Eight lines, eight **If** blocks: rows, columns, both diagonals. Wrong cell names in one line means that line never wins.
- **What the finished game does not do:** it does not stop play after a win (the other player can still place marks; the winner message stays), and it does not announce a draw when the board is full. Both are good extension tasks.

- **Space** runs **Leave the Game**, which ends this computer's session cleanly. For a clean rematch, students press H and J again; resetting the cells and winner is the tutorial's "try this next".

> **Reference projects.** Download the checkpoint projects from the wiki (10_file_exchange_multiplayer_checkpoints.zip): one project for the end of each phase, built exactly as the tutorial describes. Phase 2-4 projects use a plain folder name (tictactoe_files); change it to your shared folder in the Host and Join actions before using them on two computers.

Vocabulary Introduced

Term	What it means here
Host / join	The two roles: the host runs rounds; the other players join
Shared folder	The place where moves are exchanged as small files
Round	One turn, resolved by the host
Shared variable	A value written by one player and readable by all as global.name
Pass	Ending your turn without a move

Discussion Questions

- Why does this game use files instead of a live connection? (No firewall or port problems; works asynchronously.)
- What are the drawbacks? (Slower, needs a shared folder.)
- Why must both computers run the same project?
- How could two players play at different times of day with this system?

Differentiation

- **Support:** give the Phase 1 checkpoint; pair a stronger and a weaker student; provide the cell tables on paper.
- **Extension:** rematch reset; draw detection; stop play after a win; a score across games; a title screen instead of the H/J keys.

Worksheet Answer Key

Part A: 1-C, 2-B, 3-D, 4-A.

Part B: 1. Host a Game (File Exchange). 2. Set a Shared Variable (File Exchange). 3. End Turn (File Exchange). 4. Leave the Game (File Exchange). 5. The Round Resolved event.

Part C:

1. Advantage: it works through firewalls and needs no live connection (players can even play at different times). Disadvantage: a move takes about a second or more to appear, and both computers need a shared folder.
2. Round 5 is odd, so it is X's turn. The game tests whether global.round_number is odd (X) or even (O) and compares it with my_mark.
3. The mark only appears after the round has been confirmed by the host and published, which takes about a second (longer over a slow shared drive).

Parts D and E: completion and reflection.

Rubric: File-Exchange Tic-Tac-Toe

Level	What the game shows
4 - Complete	Board and marks work; host and join connect two computers; only the player on turn can move and both screens agree; a win is shown on both; the players can leave and start again
3 - Working	Two computers connect and take turns; the win check or the rematch is missing
2 - Partly there	The board works and the computers connect, but moves do not reach the other computer
1 - Started	Phase 1 board works on one computer only

End-of-Session Checklist

- [ ] Every student has a saved project (Ctrl+S)
- [ ] Every pair has connected at least once (H and J)
- [ ] Shared subfolders are cleaned up or can be reused next time
- [ ] Note who needs the checkpoint project next time